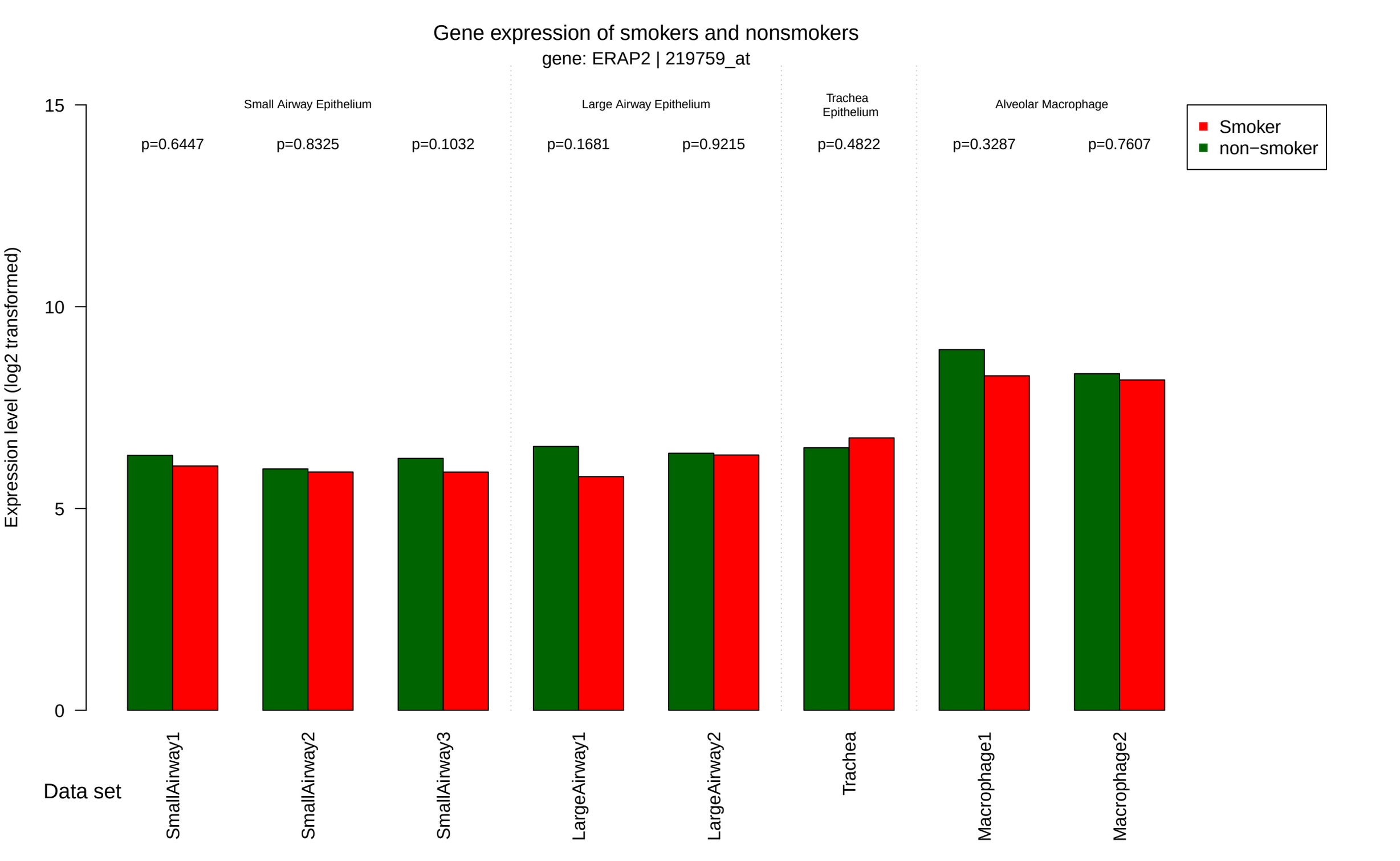
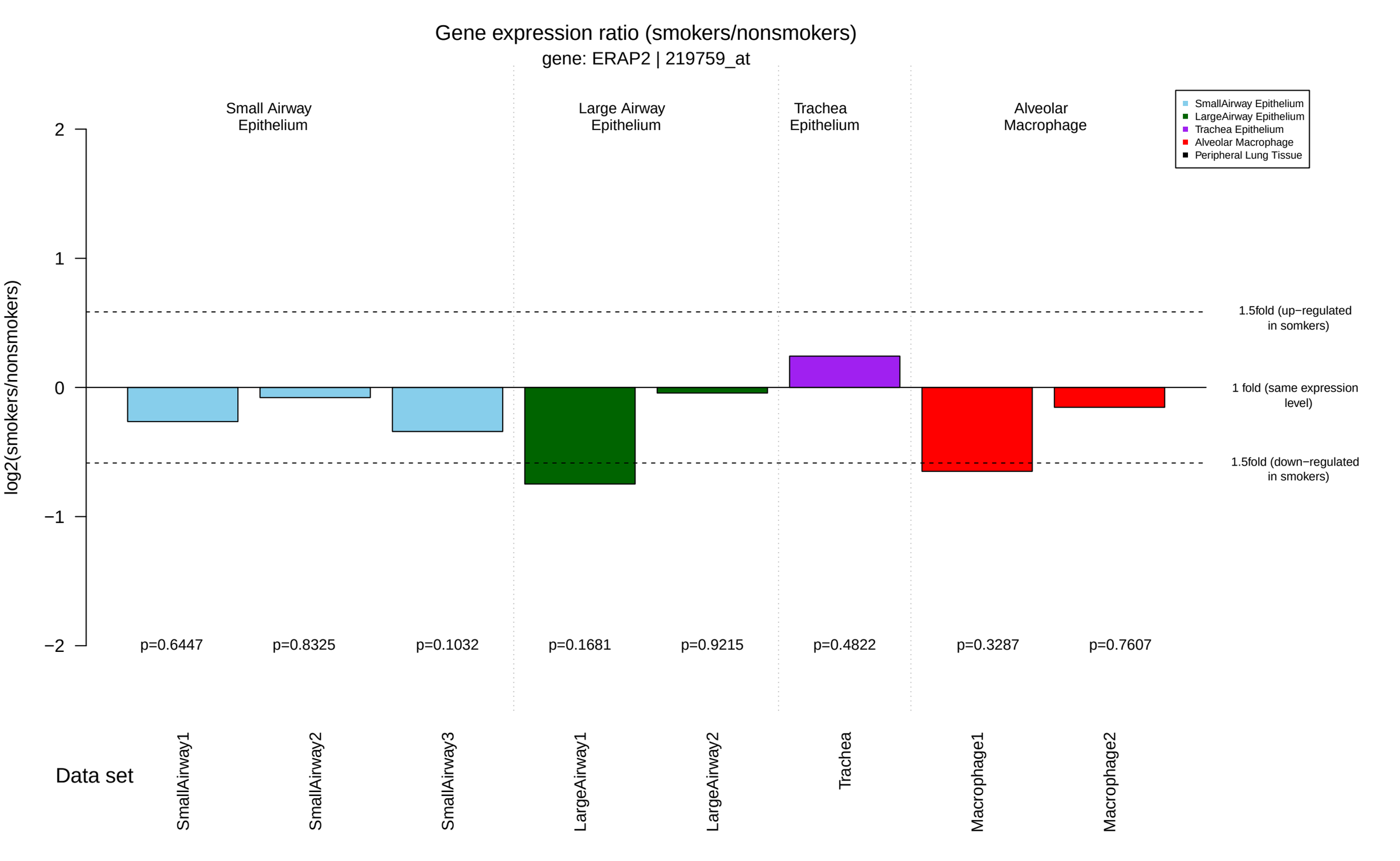
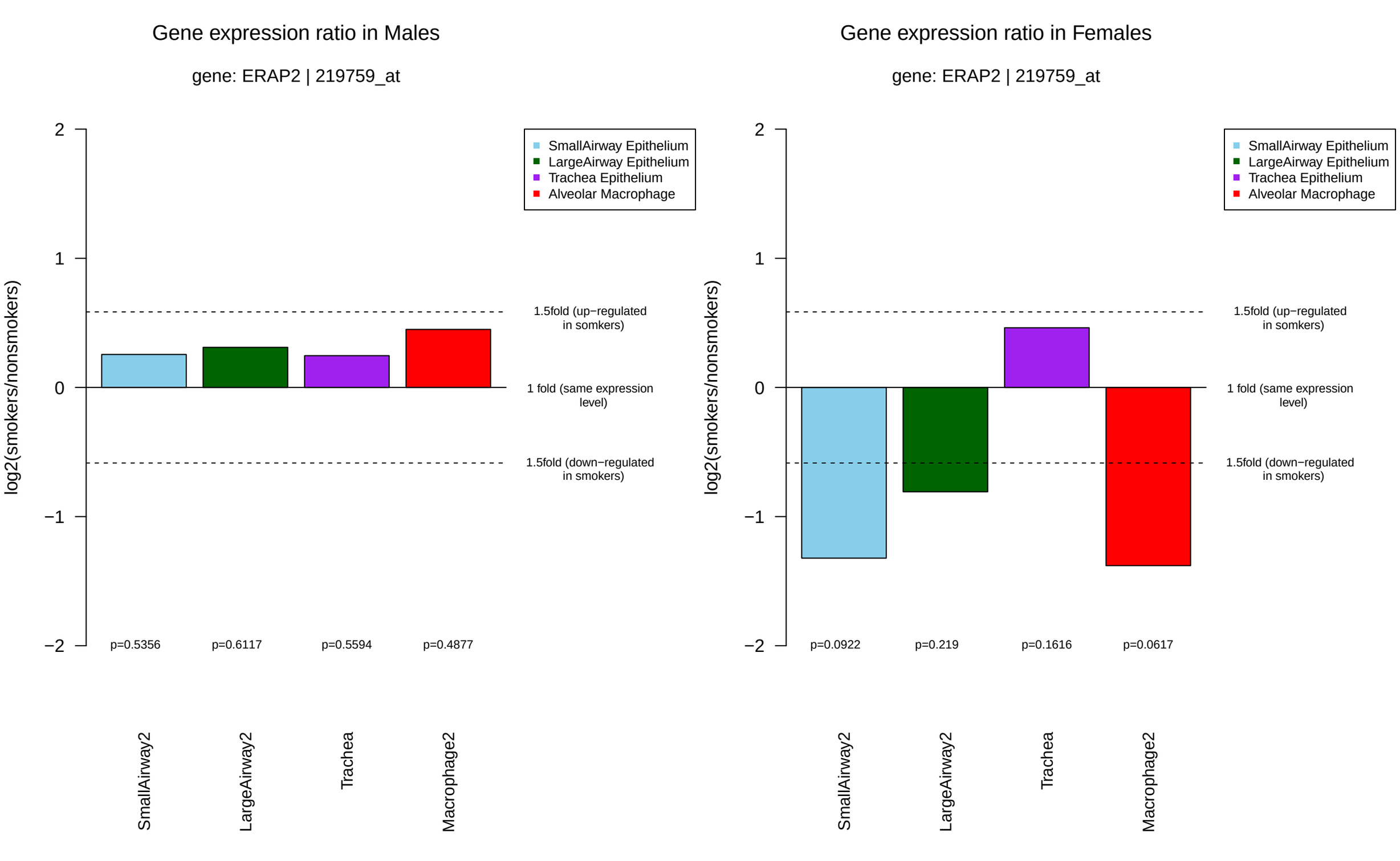
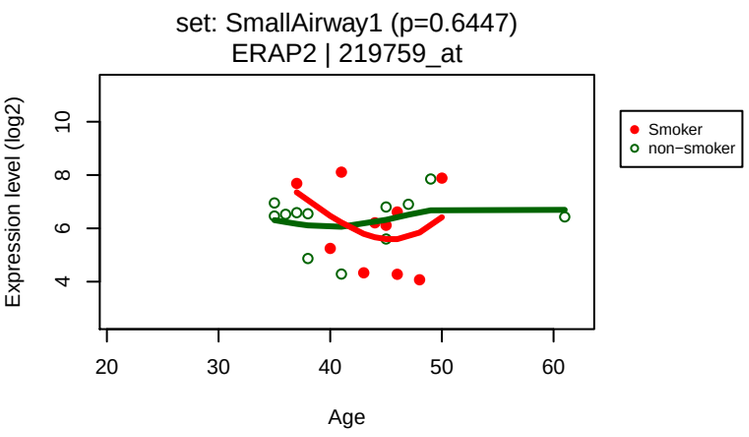
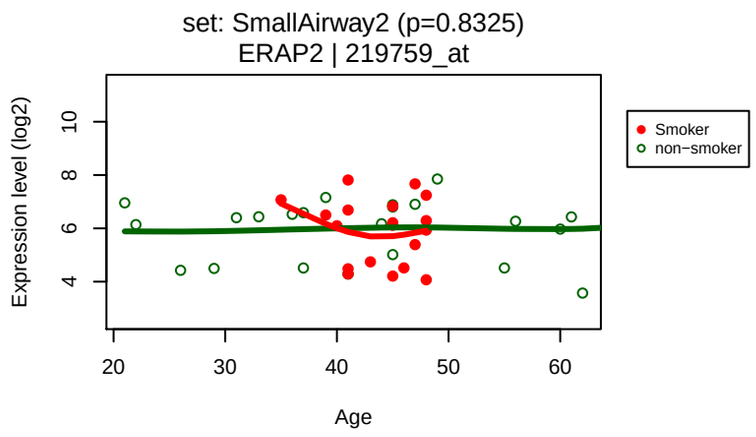
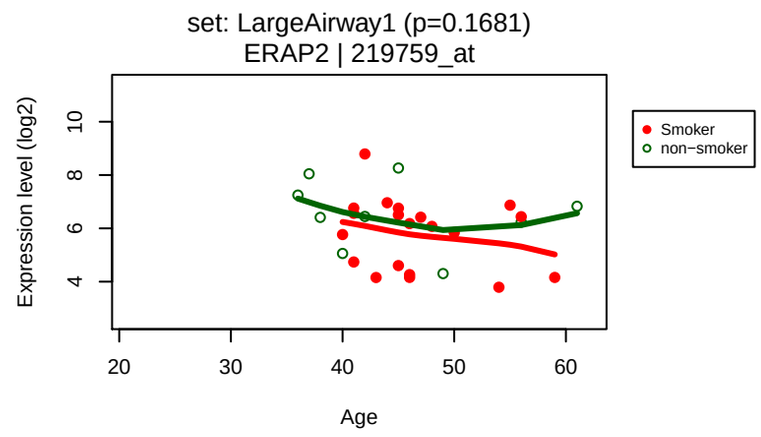

Gene expression of smokers and nonsmokers
gene: ERAP2 | 219759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6447
p=0.8325
p=0.1032
p=0.1681
p=0.9215
p=0.4822
p=0.3287
p=0.7607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

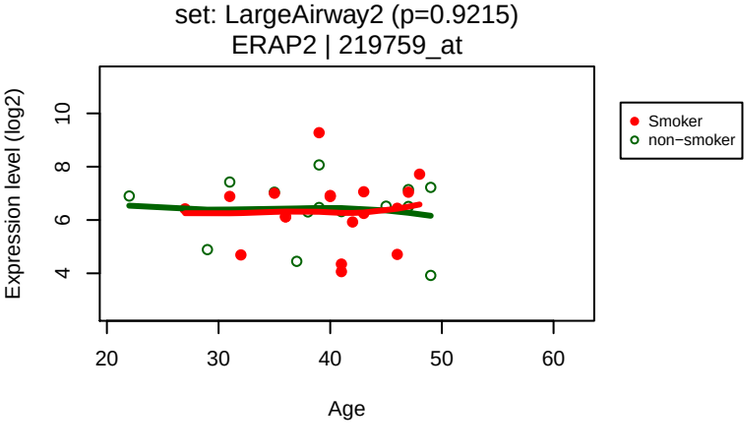

Gene expression ratio (smokers/nonsmokers)
gene: ERAP2 | 219759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6447
p=0.8325
p=0.1032
p=0.1681
p=0.9215
p=0.4822
p=0.3287
p=0.7607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

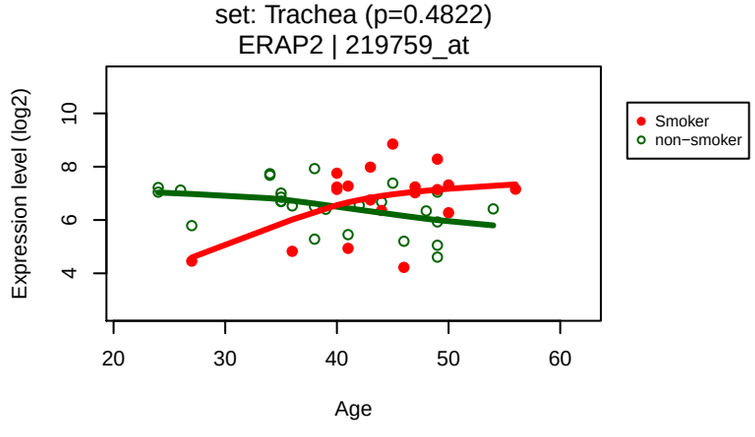

Gene expression ratio in Males
gene: ERAP2 | 219759_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5356
p=0.6117
p=0.5594
p=0.4877
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP2 | 219759_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0922
p=0.219
p=0.1616
p=0.0617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

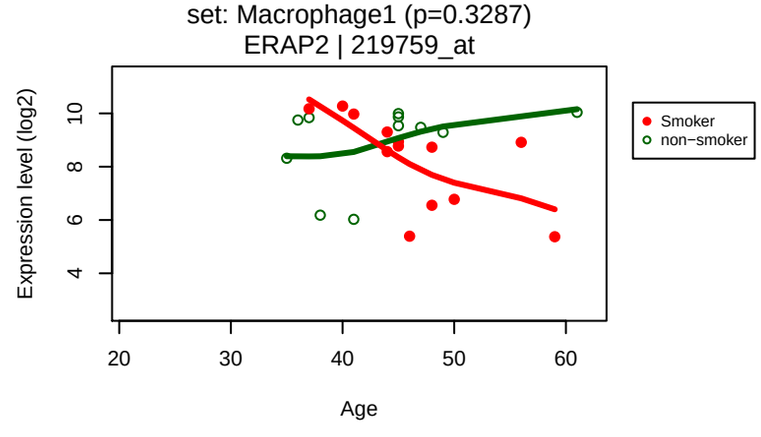

set: SmallAirway1 (p=0.6447)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker

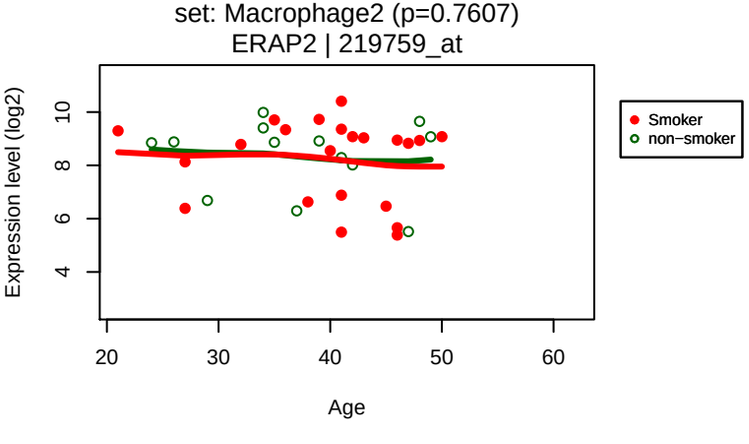

set: SmallAirway2 (p=0.8325)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker

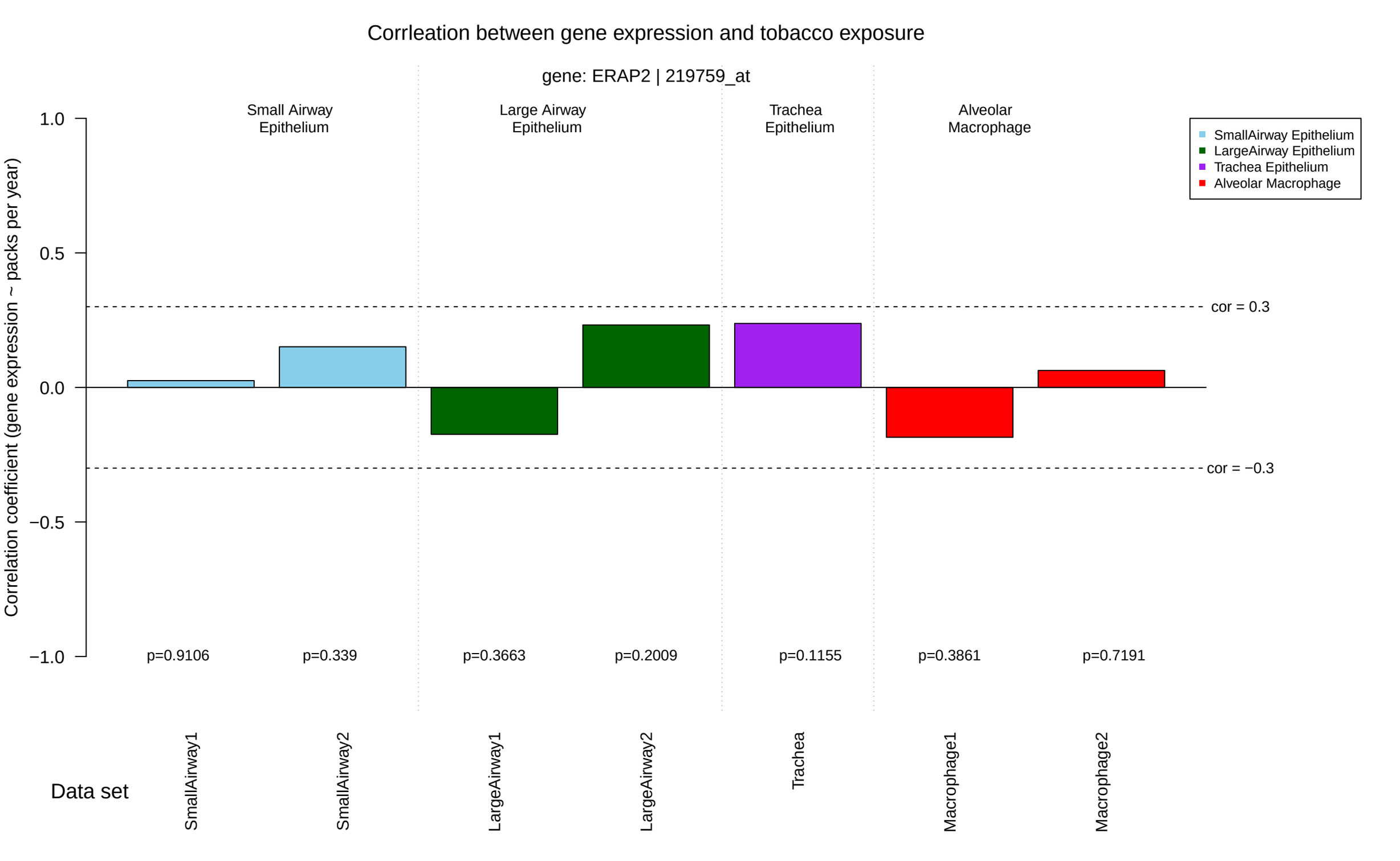

set: LargeAirway1 (p=0.1681)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9215)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4822)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3287)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7607)
ERAP2 | 219759_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP2 | 219759_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9106
p=0.339
p=0.3663
p=0.2009
p=0.1155
p=0.3861
p=0.7191
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2